

LINEAR REGRESSION

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Linear Regression in Machine learning

Linear Regression is a fundamental supervised learning algorithm used to model the relationship between a dependent variable and one or more independent variables. It predicts continuous values by fitting a straight line that best represents the data.

- It assumes that there is a linear relationship between the input and output
- Uses a best-fit line to make predictions
- Commonly used in forecasting, trend analysis, and predictive modelling



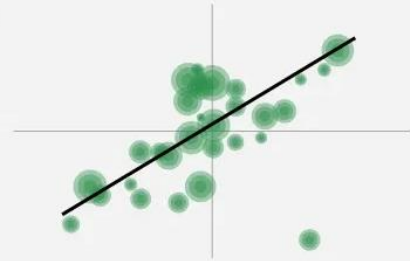
Simple model for predicting continuous values.



It finds the relationship between input features and output.



It is used in forecasting and predicting salaries based on experience.



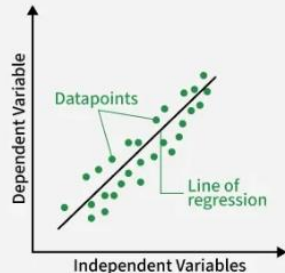
How Does Linear Regression Work?



Finds the best-fitting line by minimizing prediction errors (least squares method)



It calculates coefficients for variables that minimize the error in predictions.



For example, we want to predict a student's exam score based on how many hours they studied. We observe that as students study more hours, their scores go up. In the example of predicting exam scores based on hours studied. Here

•**Independent variable (input):** Hours studied because it's the factor we control or observe.

•**Dependent variable (output):** Exam score because it depends on how many hours were studied.

We use the independent variable to predict the dependent variable.

Best Fit Line in Linear Regression

In linear regression, the best-fit line is the straight line that most accurately represents the relationship between the independent variable (input) and the dependent variable (output). It is the line that minimizes the difference between the actual data points and the predicted values from the model.

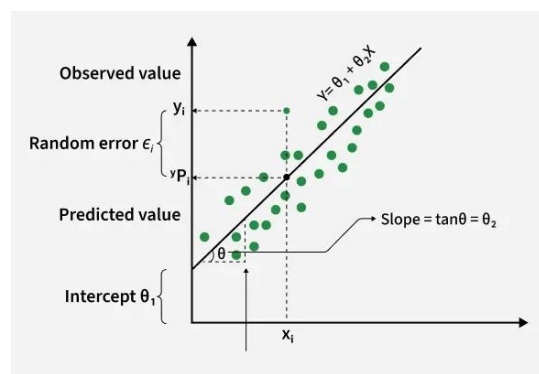
1. Goal of the Best-Fit Line

The goal of linear regression is to find a straight line that minimizes the error (the difference) between the observed data points and the predicted values. This line helps us predict the dependent variable for new, unseen data.

Here Y is called a dependent or target variable and X is called an independent variable also known as the predictor of Y .

- θ_1 represents the intercept, which is the value of Y when $X = 0$
- θ_2 represents the slope, which shows how much Y changes for a unit change in X

There are many types of functions or modules that can be used for regression. A linear function is the simplest type of function. Here, X may be a single feature or multiple features representing the problem.



2. Equation of the Best-Fit Line

For simple linear regression (with one independent variable), the best-fit line is represented by the equation

$$y = mx + b$$

Where:

- y is the predicted value (dependent variable)
- x is the input (independent variable)
- m is the slope of the line (how much y changes when x changes)
- b is the intercept (the value of y when $x = 0$)

The best-fit line will be the one that optimizes the values of m (slope) and b (intercept) so that the predicted y values are as close as possible to the actual data points.

3. Minimizing the Error: The Least Squares Method

To find the best-fit line, we use a method called Least Squares. The idea behind this method is to minimize the sum of squared differences between the actual values (data points) and the predicted values from the line. These differences are called residuals.

The formula for residuals is:

$$Residual = y_i - \hat{y}_i$$

Where:

- y_i is the actual observed value
- $\hat{y}_i$ is the predicted value from the line for that x_i

The least squares method minimizes the sum of the squared residuals:

$$\Sigma(y_i - \hat{y}_i)^2$$

This method ensures that the line best represents the data where the sum of the squared differences between the predicted values and actual values is as small as possible.

4. Interpretation of the Best-Fit Line

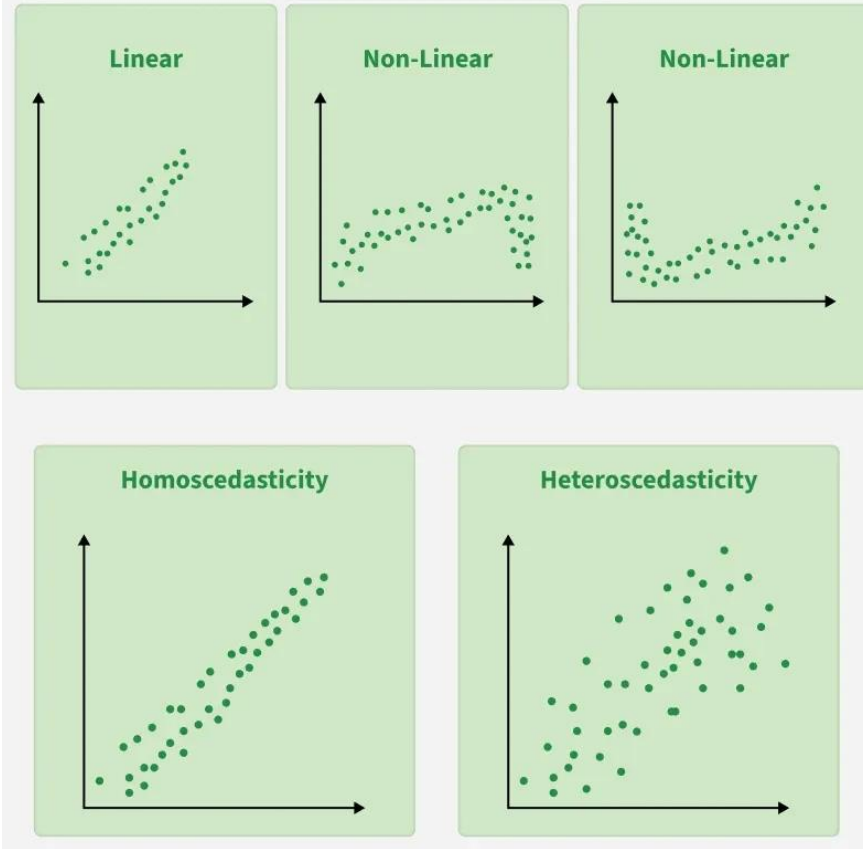
- **Slope (m):** The slope of the best-fit line indicates how much the dependent variable (y) changes with each unit change in the independent variable (x). For example, if the slope is 5, it means that for every 1-unit increase in x, the value of y increases by 5 units.
 - **Intercept (b):** The intercept represents the predicted value of y when x = 0. It's the point where the line crosses the y-axis.
- In linear regression some hypothesis are made to ensure reliability of the model's results.

Limitations:

- **Assumes Linearity:** *The method assumes the relationship between the variables is linear. If the relationship is non-linear, linear regression might not work well.*
- **Sensitivity to Outliers:** *Outliers can significantly affect the slope and intercept, skewing the best-fit line.*

Assumptions of the Linear Regression

- 1. Linearity:** The relationship between inputs (X) and the output (Y) is a straight line.
- 2. Independence of Errors:** The errors in predictions should not affect each other.
- 3. Constant Variance (Homoscedasticity):** The errors should have equal spread across all values of the input. If the spread changes (like fans out or shrinks), it's called heteroscedasticity and it's a problem for the model.
- 4. Normality of Errors:** The errors should follow a normal (bell-shaped) distribution.
- 5. No Multicollinearity (for multiple regression):** Input variables shouldn't be too closely related to each other.
- 6. No Autocorrelation:** Errors shouldn't show repeating patterns, especially in time-based data.
- 7. Additivity:** The total effect on Y is just the sum of effects from each X, no mixing or interaction between them.'



Types of Linear Regression

When there is only one independent feature it is known as Simple Linear Regression or Univariate Linear Regression and when there are more than one feature it is known as Multiple Linear Regression or Multivariate Regression.

1. Simple Linear Regression

Simple linear regression is used when we want to predict a target value (dependent variable) using only one input feature (independent variable). It assumes a straight-line relationship between the two.

Formula

$$\hat{y} = \theta_0 + \theta_1 x$$

Where:

- $\hat{y}$ is the predicted value
- x is the input (independent variable)
- θ_0 is the intercept (value of $\hat{y}$ when $x=0$)
- θ_1 is the slope or coefficient (how much $\hat{y}$ changes with one unit of x)

2. Multiple Linear Regression

Multiple linear regression involves more than one independent variable and one dependent variable. The equation for multiple linear regression is:

$$\hat{y} = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_n x_n$$

where:

- $\hat{y}$ is the predicted value
- $x_1, x_2, \dots, x_n$ are the independent variables
- $\theta_1, \theta_2, \dots, \theta_n$ are the coefficients (weights) corresponding to each predictor.
- θ_0 is the intercept.

The goal is to find the best-fit line that predicts Y accurately for given inputs X .

➤ Cost function for Linear Regression

In Linear Regression, the cost function measures how far the predicted values ($\hat{Y}$) are from the actual values (Y). It helps identify and reduce errors to find the best-fit line. The most common cost function used is Mean Squared Error (MSE), which calculates the average of squared differences between actual and predicted values:

$$\text{Cost function}(J) = \frac{1}{n} \sum_n^i (\hat{y}_i - y_i)^2$$

Here:

$$\bullet \hat{y}_i = \theta_1 + \theta_2 x_i$$

To minimize this cost, we use Gradient Descent, which iteratively updates θ_1 and θ_2 until the MSE reaches its lowest value. This ensures the line fits the data as accurately as possible.

Why Linear Regression is Important?

Here's why linear regression is important:

- It's easy to understand and interpret, making it a starting point for learning about machine learning.
- Helps predict future outcomes based on past data, making it useful in various fields like finance, healthcare and marketing.
- Many advanced algorithms, like logistic regression or neural networks, build on the concepts of linear regression.
- It's computationally efficient and works well for problems with a linear relationship.
- It's one of the most widely used techniques in both statistics and machine learning for regression tasks.
- It provides insights into relationships between variables (e.g., how much one variable influences another).

Advantages

- Simple and easy to implement with interpretable coefficients.
- Computationally efficient, suitable for large datasets and real-time applications.
- Serves as a good baseline model.
- Relatively robust to outliers.
- Widely available in ML libraries and software.

Limitations

- Assumes a linear relationship between variables.
- Sensitive to multicollinearity among features.
- Requires proper feature engineering.
- Can overfit or underfit depending on data.
- Limited for modeling complex relationships.

Simple Linear Regression math's

The linear regression line equation is written in the form

$$y = mx + c$$

where,

- *x is an Independent Variable, Plotted along the X-axis*
- *y is the Dependent Variable, Plotted along the Y-axis*

The intercept value, c, and the slope of the line, m, are evaluated using the formulas given below:

$$m = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2}$$

$$c = \frac{1}{n}(\sum y - m\sum x)$$

Solved Example Questions on Simple Linear Regression

Question 1: Find the linear regression equation for the given data:

x	y
3	8
9	6
5	4
3	2

Using formula,

$$a = \frac{\sum y \sum x^2 - \sum x \sum xy}{n(\sum x^2) - (\sum x)^2}$$

$$a = \{20 (124) - 20 (104)\} / \{4 (124) - 400\}$$
$$a = 400/96 = 4.17$$

$$b = \frac{n \sum xy - (\sum x)(\sum y)}{n \sum x^2 - (\sum x)^2}$$

Solution: Calculating intercept and slope value.

x	y	x ²	xy
3	8	9	24
9	6	81	54
5	4	25	20
3	2	9	6
$\sum x = 20$	$\sum y = 20$	$\sum x^2 = 124$	$\sum xy = 104$

$$b = \{4 (104) - 20 (20)\} / \{4 (124) - 400\}$$
$$b = 16/96 = 0.166$$

So, linear regression equation is, $y=a+bx \rightarrow y = 4.17 + 0.167x$

H.W.

1. What will be the value of y if x is 7?
2. Analysis the cost of this problem.

$$\text{Cost function}(J) = \frac{1}{n} \sum_n^i (\hat{y}_i - y_i)^2$$

Question 2: Find the linear regression equation for the given data:

x	y
4	6
7	5
3	8
1	3

Using formula,

$$a = \frac{\sum y \sum x^2 - \sum x \sum xy}{n(\sum x^2) - (\sum x)^2}$$

$$= (225 - (75) 4) / ((86) 15 - (75) 22) = 360/75$$

$$= 4.8$$

$$b = \frac{n \sum xy - (\sum x)(\sum y)}{n \sum x^2 - (\sum x)^2}$$

$$(225 - (75) 4) / ((22) 15 - (86) 4) =$$

$$= 14 / 75$$

$$= 0.1867$$

So, the linear regression equation is, $y = 4.8 + 0.187x$.

H.W.

1. What will be the value of y if x is 7?
2. Analysis the cost of this problem.

$$\text{Cost function}(J) = \frac{1}{n} \sum_n^i (\hat{y}_i - y_i)^2$$

Solution: Calculating intercept and slope value.

x	y	x ²	xy
4	6	16	24
7	5	49	35
3	8	9	24
1	3	1	3
$\sum x = 15$	$\sum y = 22$	$\sum x^2 = 75$	$\sum xy = 86$

Question 3: Find the intercept of the linear regression line if $\sum x = 25$, $\sum y = 20$, $\sum x^2 = 90$, $\sum xy = 150$, and $n = 5$.

Question 4: Find the intercept of the linear regression line if $\sum x = 30$, $\sum y = 27$, $\sum x^2 = 110$, $\sum xy = 190$, and $n = 4$.

Question 5: Find the slope of the linear regression line if $\sum x = 10$, $\sum y = 16$, $\sum x^2 = 60$, $\sum xy = 120$, and $n = 4$.

Question 6: Find the slope of the linear regression line if $\sum x = 40$, $\sum y = 32$, $\sum x^2 = 130$, $\sum xy = 210$, and $n = 4$.

Question 7: Find the slope of the linear regression line if $\sum x = 50$, $\sum y = 44$, $\sum x^2 = 150$, $\sum xy = 230$, and $n = 4$.

Multiple Linear Regression

Multiple Linear Regression extends this concept by modelling the relationship between a dependent variable and two or more independent variables. This technique allows us to understand how multiple features collectively affect the outcomes.

Steps for Multiple Linear Regression

Steps to perform multiple linear regression are similar to that of simple linear Regression but difference comes in the evaluation process. We can use it to find out which factor has the highest influence on the predicted output and how different variables are related to each other.

Equation for multiple linear regression is:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

Where:

- y is the dependent variable
- $X_1, X_2, \dots, X_n$ are the independent variables
- β_0 is the intercept
- $\beta_1, \beta_2, \dots, \beta_n$ are the slopes

The multiple regression of two variables x_1 and x_2 is given as follows:

$$y = f(x_1, x_2)$$

$$y = a_0 + a_1 x_1 + a_2 x_2$$

In general, this is given for ' n ' independent variables as:

$$y = f(x_1, x_2, \dots, x_n)$$

$$y = a_0 + a_1 x_1 + a_2 x_2 + \dots + a_n x_n + \varepsilon$$

Here,

$x_1, x_2, \dots, x_n$ are predictor (independent) variables, y is the dependent variable, $(a_0, a_1, a_2, \dots, a_n)$ are the coefficients of the regression equation, and ε is the error term.

Ex-1: Multiple Linear Regression

x1 Product 1 Sales	x2 Product 2 Sales	Y Weekly Sales
1	4	1
2	5	6
3	8	8
4	2	12

Here, the matrices for Y and X are given as

follows:

$$X = \begin{pmatrix} 1 & 1 & 4 \\ 1 & 2 & 5 \\ 1 & 3 & 8 \\ 1 & 4 & 2 \end{pmatrix} \text{ and } Y = \begin{pmatrix} 1 \\ 6 \\ 8 \\ 12 \end{pmatrix}$$

The coefficient of the multiple regression

equation is given as

$$a = \begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix}$$

The regression coefficient for multiple regression is calculated the same way as linear regression:

$$\hat{a} = ((X^T X)^{-1} X^T) Y$$

$$X^T X = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 5 & 8 & 2 \end{pmatrix} \begin{pmatrix} 1 & 1 & 4 \\ 1 & 2 & 5 \\ 1 & 3 & 8 \\ 1 & 4 & 2 \end{pmatrix} = \begin{pmatrix} 4 & 10 & 19 \\ 10 & 30 & 46 \\ 19 & 46 & 109 \end{pmatrix}$$

$$(X^T X)^{-1} = \begin{pmatrix} 4 & 10 & 19 \\ 10 & 30 & 46 \\ 19 & 46 & 109 \end{pmatrix}^{-1} = \begin{pmatrix} 3.15 & -0.59 & -0.30 \\ -0.59 & 0.20 & 0.016 \\ -0.30 & 0.016 & 0.054 \end{pmatrix}$$

$$X^T X)^{-1} X^T = \begin{pmatrix} 3.15 & -0.59 & -0.30 \\ -0.59 & 0.20 & 0.016 \\ -0.30 & 0.016 & 0.054 \end{pmatrix} \times \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 5 & 8 & 2 \end{pmatrix} = \begin{pmatrix} 0.05 & 0.47 & -1.02 & 0.19 \\ -0.32 & -0.098 & 0.155 & 0.26 \\ -0.065 & 0.005 & 0.185 & -0.125 \end{pmatrix}$$

$$\hat{a} = ((X^T X)^{-1} X^T) Y = \begin{pmatrix} 0.05 & 0.47 & -1.02 & 0.19 \\ -0.32 & -0.098 & 0.155 & 0.26 \\ -0.065 & 0.005 & 0.185 & -0.125 \end{pmatrix} \times \begin{pmatrix} 1 \\ 6 \\ 8 \\ 12 \end{pmatrix} = \begin{pmatrix} -1.69 \\ 3.48 \\ -0.05 \end{pmatrix}$$

$$a_0 = -1.69$$

$$a_1 = 3.48$$

$$a_2 = -0.05$$

- $y = a_0 + a_1 x_1 + a_2 x_2$
- Hence, the constructed model is:
- $y = -1.69 + 3.48 x_1 - 0.05 x_2$

Ex-2: Multiple Linear Regression

Y	X1	X2
78.5	7	2.6
74.3	1	2.9
104.3	11	5.6
87.6	11	3.1
95.9	7	5.2
109.2	11	5.5
102.7	3	7.1

The data in the table relate grams plant dry weight y, to present soil organic matter x1, and kilograms of supplemental soil nitrogen added per 100 square meters x2.

Q1. Obtain the regression equation.

Q2. Also, predict the dry weight given soil organic matter of 5 and soil nitrogen 4.

Multiple regression equation is given by,

$$y = \beta_0 + \beta_1 * x_1 + \beta_2 * x_2$$

Where, β_0, β_1 and β_2 are regression coefficients of multiple regression.

$$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix}$$

$$\beta = (X^T X)^{-1} X^T Y$$

$$Y = \begin{bmatrix} 78.5 \\ 74.3 \\ 104.3 \\ 87.6 \\ 95.6 \\ 109.2 \\ 102.7 \end{bmatrix} \quad X = \begin{bmatrix} 1 & 7 & 2.6 \\ 1 & 1 & 2.9 \\ 1 & 11 & 5.6 \\ 1 & 11 & 3.1 \\ 1 & 7 & 5.2 \\ 1 & 11 & 5.5 \\ 1 & 3 & 7.1 \end{bmatrix}$$

$$Y = \begin{bmatrix} 78.5 \\ 74.3 \\ 104.3 \\ 87.6 \\ 95.9 \\ 109.2 \\ 102.7 \end{bmatrix} \quad X = \begin{bmatrix} 1 & 7 & 2.6 \\ 1 & 1 & 2.9 \\ 1 & 11 & 5.6 \\ 1 & 11 & 3.1 \\ 1 & 7 & 5.2 \\ 1 & 11 & 5.5 \\ 1 & 3 & 7.1 \end{bmatrix} \quad X^T = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 7 & 1 & 11 & 11 & 7 & 11 & 3 \\ 2.6 & 2.9 & 5.6 & 3.1 & 5.2 & 5.5 & 7.1 \end{bmatrix}$$

$$X^T * X = \begin{bmatrix} 7 & 51 & 32 \\ 51 & 471 & 235 \\ 32 & 235 & 163.84 \end{bmatrix} \quad (X^T * X)^{-1} = \begin{bmatrix} 1.7996 & -0.0685 & -0.2532 \\ -0.0685 & -0.0101 & -0.0011 \\ -0.2532 & -0.0011 & 0.0571 \end{bmatrix}$$

$$\beta = (X^T X)^{-1} X^T Y$$

$$X^T = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 7 & 1 & 11 & 11 & 7 & 11 & 3 \\ 2.6 & 2.9 & 5.6 & 3.1 & 5.2 & 5.5 & 7.1 \end{bmatrix}$$

$$(X^T * X)^{-1} = \begin{bmatrix} 1.7996 & -0.0685 & -0.2532 \\ -0.0685 & 0.0101 & -0.0011 \\ -0.2532 & -0.0011 & 0.0571 \end{bmatrix}$$

$$(X^T * X)^{-1} X^T = \begin{bmatrix} 0.66178 & 0.99682 & -0.37182 & 0.26118 & 0.00346 & -0.3465 & -0.20362 \\ -0.00066 & -0.06159 & 0.03644 & 0.03919 & -0.00352 & 0.03655 & -0.04601 \\ -0.11244 & -0.08871 & 0.05446 & -0.08829 & 0.03602 & 0.04875 & 0.14891 \end{bmatrix}$$

$$(X^T * X)^{-1} X^T = \begin{bmatrix} 0.66178 & 0.99682 & -0.37182 & 0.26118 & 0.00346 & -0.3465 & -0.20362 \\ -0.00066 & -0.06159 & 0.03644 & 0.03919 & -0.00352 & 0.03655 & -0.04601 \\ -0.11244 & -0.08871 & 0.05446 & -0.08829 & 0.03602 & 0.04875 & 0.14891 \end{bmatrix} \quad Y = \begin{bmatrix} 78.5 \\ 74.3 \\ 104.3 \\ 87.6 \\ 95.9 \\ 109.2 \\ 102.7 \end{bmatrix}$$

$$\beta = (X^T * X)^{-1} X^T Y = \begin{bmatrix} 51.9 \\ 1.5 \\ 6.5 \end{bmatrix}$$

$$\beta = (X^T * X)^{-1} X^T Y = \begin{bmatrix} 51.9 \\ 1.5 \\ 6.5 \end{bmatrix} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix}$$

$$y = \beta_0 + \beta_1 * x_1 + \beta_2 * x_2$$

$$y = 51.9 + 1.5 * x_1 + 6.5 * x_2$$

Y is the Obtain the regression equation.

Q2. Also, predict the dry weight given soil organic matter of 5 and soil nitrogen 4.

$$y = 51.9 + 1.5 * 5 + 6.5 * 4$$

$$y = 85.4$$

Ex-2: Multiple Linear Regression: Let's create a realistic, small-scale dataset for our housing example. We'll use three independent variables to keep the manual calculation manageable, but the matrix formula works for any number.

Data Collected:

S.No:	Size(x_1)	Floor(x_2)	Broadband(x_3)	Rental Price(y)
1	500	4	8	320
2	550	7	50	380
3	620	9	7	400
4	630	5	24	390
5	665	8	100	385
6	700	4	8	410
7	770	10	7	480
8	880	12	50	600
9	920	14	8	570
10	1000	9	24	620

Let's build $\mathbf{X}$ matrix (with a column of 1s for the intercept) and $\mathbf{y}$ vector.

$$\mathbf{X} = \begin{bmatrix} 1 & 500 & 4 & 8 \\ 1 & 550 & 7 & 50 \\ 1 & 620 & 9 & 7 \\ 1 & 630 & 5 & 24 \\ 1 & 665 & 8 & 100 \\ 1 & 700 & 4 & 8 \\ 1 & 770 & 10 & 7 \\ 1 & 880 & 12 & 50 \\ 1 & 920 & 14 & 8 \\ 1 & 1000 & 9 & 24 \end{bmatrix} \quad \mathbf{y} = \begin{bmatrix} 320 \\ 380 \\ 400 \\ 390 \\ 385 \\ 410 \\ 480 \\ 600 \\ 570 \\ 620 \end{bmatrix}$$

Now compute $\mathbf{X}^\top \mathbf{X}$, $\mathbf{X}^\top \mathbf{y}$, and $(\mathbf{X}^\top \mathbf{X})^{-1}$

$$\mathbf{X}^\top = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 500 & 550 & 620 & 630 & 665 & 700 & 770 & 880 & 920 & 1000 \\ 4 & 7 & 9 & 5 & 8 & 4 & 10 & 12 & 14 & 9 \\ 8 & 50 & 7 & 24 & 100 & 8 & 7 & 50 & 8 & 24 \end{bmatrix}$$

$$\mathbf{X}^\top \mathbf{X} = \begin{bmatrix} n & \sum x_1 & \sum x_2 & \sum x_3 \\ \sum x_1 & \sum x_1^2 & \sum x_1 x_2 & \sum x_1 x_3 \\ \sum x_2 & \sum x_1 x_2 & \sum x_2^2 & \sum x_2 x_3 \\ \sum x_3 & \sum x_1 x_3 & \sum x_2 x_3 & \sum x_3^2 \end{bmatrix}$$

$$\mathbf{X}^\top \mathbf{X} = \begin{bmatrix} 10 & 7235 & 82 & 286 \\ 7235 & 5479725 & 62840 & 203810 \\ 82 & 62840 & 772 & 2395 \\ 286 & 203810 & 2395 & 16442 \end{bmatrix}$$

$$\mathbf{X}^\top \mathbf{y} = \begin{bmatrix} \sum y \\ \sum x_1 y \\ \sum x_2 y \\ \sum x_3 y \end{bmatrix}$$

$$(\mathbf{X}^\top \mathbf{X})^{-1} = \begin{bmatrix} 2.549 & -0.004 & 0.054 & -0.005 \\ -0.004 & 0.000 & -0.000 & 0.000 \\ 0.054 & -0.000 & 0.021 & -0.000 \\ -0.005 & 0.000 & -0.000 & 0.000 \end{bmatrix}$$

$$\beta = (\mathbf{X}^\top \mathbf{X})^{-1} \cdot \mathbf{X}^\top \mathbf{y} = \begin{bmatrix} 19.56 \\ 0.55 \\ 4.96 \\ -0.06 \end{bmatrix}$$

After getting the values of coefficients (β s), the estimated multiple linear regression model becomes:

$$\text{Predicted Rent} = 19.56 + 0.55 \times (\text{Size}) + 4.96 \times (\text{Floor}) - 0.06 \times (\text{Broadband})$$

Now make prediction on above model and compute the sum of squared error.

$$\text{Cost function}(J) = \frac{1}{n} \sum_n^i (\hat{y}_i - y_i)^2$$

S.No:	Size(Sqft)	Actual Prize(y)	Predicted Prize($\hat{y}$)	$(y - \hat{y})^2 = \epsilon^2$
1	500	320	313.92	36.97
2	550	380	353.78	687.49
3	620	400	404.78	22.85
4	630	390	389.42	0.34
5	665	385	418.99	1155.32
6	700	410	423.92	193.77
7	770	480	492.24	149.82
8	880	600	560.08	1593.61
9	920	570	594.52	601.23
10	1000	620	612.76	52.42

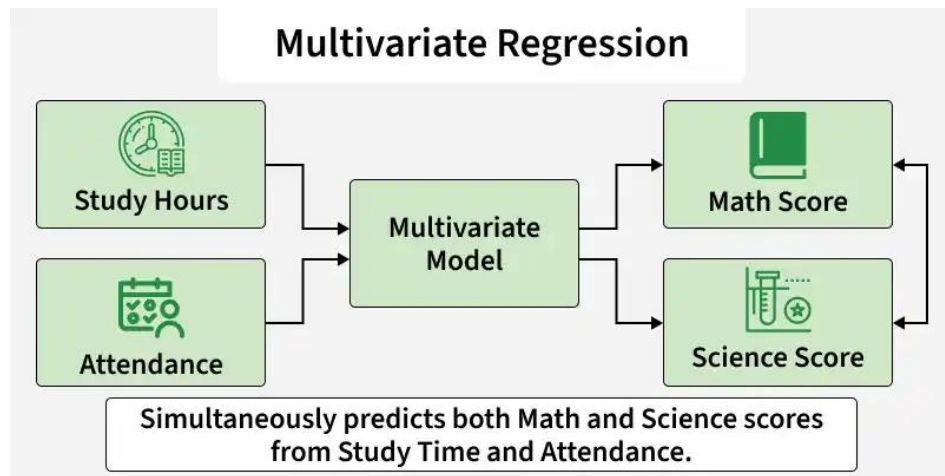
$$\text{Sum of Squared Error} = \text{SSE} = \sum_i \epsilon_i^2 = 4493.82/10 = ??$$

Multivariate linear regression

Multivariate linear regression, finds the relationship between one independent variable and multiple number of dependent variables. This model is similar to multiple regression solving the normal equations and estimate the regression coefficients.

Multivariate Regression is a technique used when we need to predict more than one output variable at the same time. Instead of building separate models for each target, a single model learns how input features are connected to multiple outputs together. This is especially useful when the outputs are related to each other and may influence one another.

- Gives two or more results at the same time.
- One model handles all target variables together.
- Use multiple features to make predictions.
- If targets are connected, the model learns them better together.



Multiple Linear Regression vs Multivariate Regression

Features	Multiple Linear Regression	Multivariate Regression
Outputs	One dependent variable	More than one dependent variable
Inputs	Multiple inputs allowed	Multiple inputs allowed
Model Structure	Separate model for each output	Single combined model
Output Relationship	Does not consider relation between outputs	Learns relationships among outputs
Coefficient Form	Coefficient vector	Coefficient matrix

Suppose the number of dependent variables are m , and the number of observations are n . The general form of multivariate linear regression is given by,

$$y_i = \beta_{0i} + \beta_{1i}x_i$$

Where,

$$i = 1, 2, 3, \dots, m$$

β_{0i} and β_{1i} are the regression coefficients of i th dependent variables.

Ex-1: apply multivariate linear regression

X_0	Y_1	Y_2
0	1	-1
1	4	-1
2	3	2
3	8	3
4	9	2

$$y_i = \beta_{0i} + \beta_{1i}x_0$$

$$y_1 = \beta_{01} + \beta_{11}x_0$$

$$y_2 = \beta_{02} + \beta_{12}x_0$$

$$\beta = \begin{bmatrix} \beta_{01} & \beta_{02} \\ \beta_{11} & \beta_{12} \end{bmatrix}$$

$$\beta = (X^T X)^{-1} X^T Y$$

$$X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix} \quad Y = \begin{bmatrix} 1 & -1 \\ 4 & -1 \\ 3 & 2 \\ 8 & 3 \\ 9 & 2 \end{bmatrix}$$

$$X^T = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 & 4 \end{bmatrix}$$

$$(X^T X) = \begin{bmatrix} 5 & 10 \\ 10 & 30 \end{bmatrix}$$

$$(X^T X)^{-1} = \begin{bmatrix} 0.6 & -0.25 \\ -0.2 & 0.1 \end{bmatrix}$$

$$\beta = (X^T X)^{-1} X^T Y = \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}$$

$$y_1 = \beta_{01} + \beta_{11}x_0$$

$$y_1 = 1 + 2x_0$$

$$y_2 = \beta_{02} + \beta_{12}x_0$$

$$y_2 = -1 + x_0$$

Polynomial Regression

If your data points clearly will not fit a linear regression (a straight line through all data points), it might be ideal for polynomial regression. Polynomial regression, like linear regression, uses the relationship between the variables x and y to find the best way to draw a line through the data points.

Polynomial Regression is a form of linear regression where the relationship between the independent variable (x) and the dependent variable (y) is modelled as an n^{th} degree polynomial. It is useful when the data exhibits a non-linear relationship allowing the model to fit a curve to the data.

How does a Polynomial Regression work?

Polynomial regression is an extension of [linear regression](#) where higher-degree terms are added to model non-linear relationships. The general form of the equation for a polynomial regression of degree n is:

$$y = \beta_0 + \beta_1 x^1 + \beta_2 x^2 + \dots + \beta_n x^n + \epsilon$$

where:

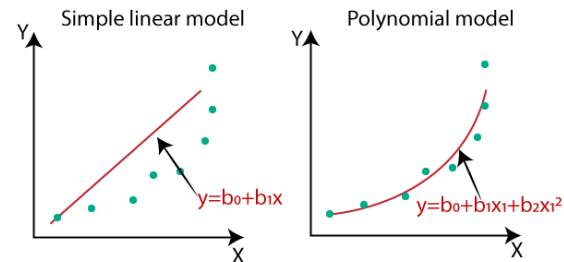
- y is the dependent variable.
- x is the independent variable.
- $\beta_0, \beta_1, \dots, \beta_n$ are the coefficients of the polynomial terms.
- n is the degree of the polynomial.
- ϵ represents the error term.

❑ Why do we need Polynomial Regression?

Let's consider a case of Simple Linear Regression.

We make our model and find out that it performs very badly. We observe between the actual value and the best fit line, which we predicted and it seems that the actual value has some kind of curve in the graph and our line is no where near to cutting the mean of the points. Polynomial Regression does not require the relationship between the independent and dependent variables to be linear in the dataset. This is also one of the main difference between the Linear and Polynomial Regression. Polynomial Regression is generally used when the points in the data are not captured by the Linear Regression Model and the Linear Regression fails in describing the best result clearly.

This where polynomial Regression comes to the play, it predicts the best fit line that follows the pattern(curve) of the data, as shown in the pic below:



So, the need for Polynomial Regression is:-

- Non-linear Relationships:** Polynomial regression is used when the relationship between the independent variable (input) and dependent variable (output) is non-linear. Unlike linear regression which fits a straight line, it fits a polynomial equation to capture the curve in the data.
- Better Fit for Curved Data:** When a researcher hypothesizes a curvilinear relationship, polynomial terms are added to the model. A linear model often results in residuals with noticeable patterns which shows a poor fit. It can capture these non-linear patterns effectively.
- Flexibility and Complexity:** It does not assume all independent variables are independent. By introducing higher-degree terms, it allows for more flexibility and can model more complex, curvilinear relationships between variables.

❑ Balancing Overfitting and Underfitting in Polynomial Regression

In polynomial regression, overfitting happens when the model is too complex and fits the training data too closely helps in making it perform poorly on new data. To avoid this, we use techniques like Lasso and Ridge regression which helps to simplify the model by limiting the size of the coefficients.

On the other hand, underfitting occurs when the model is too simple to capture the real patterns in the data. This usually happens with a low-degree polynomial. The key is to choose the right polynomial degree to ensure the model is neither too complex nor too simple which helps it work well on both the training data and new data.

❑ Disadvantages of Polynomial Regression

- 1.**Sensitivity to Outliers:** It is highly sensitive to outliers and a few extreme values can skew the results significantly.
- 2.**Overfitting:** With higher-degree polynomials, there's a risk of overfitting where the model becomes too complex and fails to generalize well to new data.
- 3.**Limited Outlier Detection:** Unlike linear regression, it has fewer built-in methods for detecting or handling outliers helps in making it challenging to identify when the model is affected by them.

Consider the polynomial of 2nd degree.

The polynomial equation is given by $y = a_0 + a_1x + a_2x^2$

The coefficients a_0, a_1 and a_2 are calculated using the formula,

$$a = X^{-1}B$$

$$X = \begin{bmatrix} n & \sum x_i & \sum x_i^2 \\ \sum x_i & \sum x_i^2 & \sum x_i^3 \\ \sum x_i^2 & \sum x_i^3 & \sum x_i^4 \end{bmatrix} \quad B = \begin{bmatrix} \sum y_i \\ \sum(x_i, y_i) \\ \sum(x_i^2, y_i) \end{bmatrix}$$

Ex-1: apply polynomial regression

x	y
1	1
2	4
3	9
4	15

x_i	y_i	$x_i y_i$	x_i^2	$x_i^2 y_i$	x_i^3	x_i^4
1	1	1	1	1	1	1
2	4	8	4	16	8	16
3	9	27	9	81	27	81
4	15	60	16	240	64	256
$\sum x_i = 10$	$\sum y_i = 29$	$\sum x_i y_i = 96$	$\sum x_i^2 = 30$	$\sum x_i^2 y_i = 338$	$\sum x_i^3 = 100$	$\sum x_i^4 = 354$

$$a = X^{-1}B \quad X = \begin{bmatrix} n & \sum x_i & \sum x_i^2 \\ \sum x_i & \sum x_i^2 & \sum x_i^3 \\ \sum x_i^2 & \sum x_i^3 & \sum x_i^4 \end{bmatrix} \quad B = \begin{bmatrix} \sum y_i \\ \sum(x_i, y_i) \\ \sum(x_i^2, y_i) \end{bmatrix} \quad \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 4 & 10 & 30 \\ 10 & 30 & 100 \\ 30 & 100 & 354 \end{bmatrix}^{-1} \times \begin{bmatrix} 29 \\ 96 \\ 338 \end{bmatrix} = \begin{pmatrix} -0.75 \\ 0.95 \\ 0.75 \end{pmatrix}$$

$$a = X^{-1}B \quad \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 4 & 10 & 30 \\ 10 & 30 & 100 \\ 30 & 100 & 354 \end{bmatrix}^{-1} \times \begin{bmatrix} 29 \\ 96 \\ 338 \end{bmatrix} = \begin{pmatrix} -0.75 \\ 0.95 \\ 0.75 \end{pmatrix} \quad y = -0.75 + 0.95x + 0.75x^2$$

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- [Linear Regression Solved Example](#)
- [Multiple Linear Regression Solved Numerical Example](#)
- [Multiple Linear Regression Solved Numerical Example](#)
- [Multivariate Linear Regression Solved Example Multiple Regression](#)
- [Polynomial Regression Algorithm and Solved Numerical Example](#)